CLASS 9 · CBSE · GANITA MANJARI PART 1 · SHORT NOTES

Chapter 1: Orienting Yourself: The Use of Coordinates

Topic-wise short notes with solved examples

Two numbers are enough to fix the position of any point in a plane. This chapter sets up the coordinate plane and uses it to find the distance between two points.

1 The Cartesian plane

- A **coordinate system** uses numbers to describe the exact position of a point.
- Two number lines are drawn at right angles. The horizontal line is the x**-axis** and the vertical line is the y**-axis**. Together they are the **coordinate axes**.
- The axes meet at the **origin** O, whose coordinates are $(0, 0)$.
- Equal units are marked on both axes. Distances to the **right** of O and **upwards** from O are positive. Distances to the **left** and **downwards** are negative.
- The plane containing the axes is called the **Cartesian plane**, the **coordinate plane** or the xy**-plane**.

REMEMBER

The system is named after René Descartes (1637). The negative half of each axis depends on Brahmagupta's work on zero and negative numbers (628 CE).

2 Coordinates of a point

- Every point P in the plane is written as $P(x, y)$.
- The x**-coordinate** is the perpendicular distance of P from the y-axis, measured along the x-axis.
- The y**-coordinate** is the perpendicular distance of P from the x-axis, measured along the y-axis.
- The order matters. If $x \neq y$, then $(x, y) \neq (y, x)$. The two are the same point only when $x = y$.
- A point on the x-axis has the form $(x, 0)$. A point on the y-axis has the form $(0, y)$.

Example 2.1 Write the coordinates of the point that is (a) on the x-axis, 4.5 units to the right of O (b) on the y-axis, 4.5 units below O.

SOLUTION

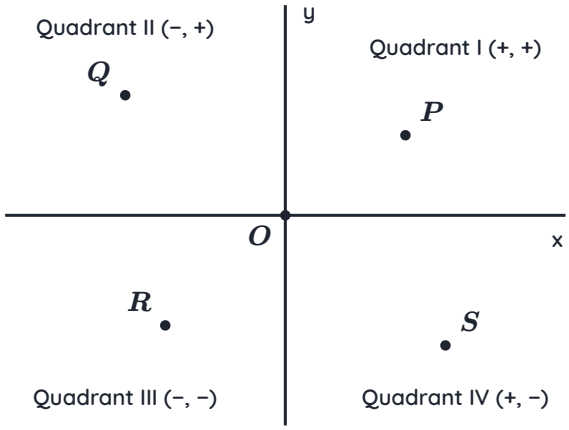
(a) It is on the x-axis, so $y = 0$. To the right means x is positive.

(b) It is on the y-axis, so $x = 0$. Below means y is negative.

(a) $(4.5, 0)$ **(b)** $(0, -4.5)$

3 The four quadrants

- The axes divide the plane into four parts called **quadrants**, numbered anticlockwise from the top right.



P(3, 2), Q(−4, 3), R(−3, −3) and S(4, −3.5) lie in the four quadrants

Quadrant	Sign of x	Sign of y	Example
I	$+$	$+$	$(3, 2)$
II	$-$	$+$	$(-5, 3)$
III	$-$	$-$	$(-3, -3)$
IV	$+$	$-$	$(3, -5)$

- A point on an axis does not belong to any quadrant.

Example 3.1 In which quadrant, or on which axis, does each point lie: $(-5, 3)$, $(3, -5)$, $(0, 4)$, $(-2.9, 0)$?

SOLUTION

$(-5, 3)$: x negative, y positive. $(3, -5)$: x positive, y negative.

$(0, 4)$ has $x = 0$ and $(-2.9, 0)$ has $y = 0$.

Quadrant II; Quadrant IV; on the y-axis; on the x-axis

4 Distance along a line parallel to an axis

- Two points with the same y-coordinate lie on a line parallel to the x-axis. The distance between (x_1, y) and (x_2, y) is $|x_2 - x_1|$.
- Two points with the same x-coordinate lie on a line parallel to the y-axis. The distance between (x, y_1) and (x, y_2) is $|y_2 - y_1|$.
- The bars mean the **absolute value**: take the difference and ignore its sign.

Example 4.1 A door has its two ends at $(8.5, 0)$ and $(11.5, 0)$. How wide is it? Another door has ends at $(0, 1.5)$ and $(0, 4)$. Which door is wider?

SOLUTION

First door: $|11.5 - 8.5| = 3$ units. Second door: $|4 - 1.5| = 2.5$ units.

The first door is 3 units wide and is the wider one.

Example 4.2 Three corners of a rectangular table are at $(8, 9)$, $(11, 9)$ and $(11, 7)$. Find the fourth corner and the length and width of the table.

SOLUTION

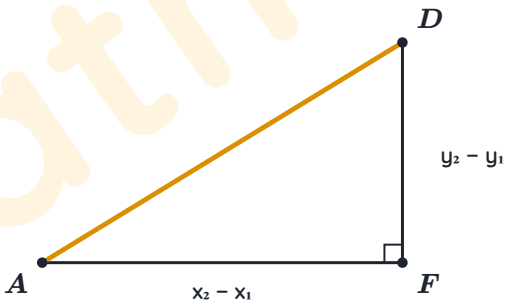
The fourth corner has the x-coordinate of the first point and the y-coordinate of the third.

Length $= |11 - 8| = 3$ units and width $= |9 - 7| = 2$ units.

Fourth corner $(8, 7)$; the table is 3 units by 2 units.

5 The distance formula

- For a slanting segment, draw a right-angled triangle with sides parallel to the axes and use the **Baudhāyana–Pythagoras theorem**.



AD is the hypotenuse of a right-angled triangle

- The distance between $A(x_1, y_1)$ and $D(x_2, y_2)$ is

$$AD = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

- It makes no difference whether $x_2 - x_1$ and $y_2 - y_1$ are positive or negative, because they are squared.
- The distance of (x, y) from the origin is $\sqrt{x^2 + y^2}$.

Example 5.1 Find the lengths of the sides of the triangle with vertices $A(3,4)$, $D(7,1)$ and $M(9,6)$.

SOLUTION

$AD = \sqrt{(7-3)^2+(1-4)^2} = \sqrt{16+9} = 5$ units

$DM = \sqrt{(9-7)^2+(6-1)^2} = \sqrt{4+25} = \sqrt{29}$ units

$MA = \sqrt{(3-9)^2+(4-6)^2} = \sqrt{36+4} = \sqrt{40}$ units

$AD = 5$, $DM = \sqrt{29}$ and $MA = \sqrt{40}$ units

Example 5.2 Show that $A(2,1)$, $B(-1,2)$, $C(-2,-1)$ and $D(1,-2)$ are the vertices of a square, and find its area.

SOLUTION

$AB^2 = 9+1 = 10$, $BC^2 = 1+9 = 10$, $CD^2 = 9+1 = 10$, $DA^2 = 1+9 = 10$, so all four sides are equal.

Diagonals: $AC^2 = 16+4 = 20$ and $BD^2 = 4+16 = 20$, so the diagonals are equal.

Four equal sides and equal diagonals make a square. Area $= (\sqrt{10})^2$.

$ABCD$ is a square of area 10 square units.

6 Reflection in the axes

- Reflecting a point in the y-**axis** changes the sign of its x-coordinate: $(x,y) \to (-x,y)$.
- Reflecting a point in the x-**axis** changes the sign of its y-coordinate: $(x,y) \to (x,-y)$.
- A reflection does not change lengths. A triangle and its image have equal sides.

Example 6.1 Triangle ADM with $A(3,4)$, $D(7,1)$, $M(9,6)$ is reflected in the y-axis. Write the images and check that $A'D' = AD$.

SOLUTION

$A'(-3,4)$, $D'(-7,1)$ and $M'(-9,6)$

$A'D' = \sqrt{(-7+3)^2+(1-4)^2} = \sqrt{16+9} = 5$ units, the same as AD.

The reflection keeps the lengths of the sides.

7 Using distances: collinear points, mid-points and circles

- **Collinear points**: three points lie on one line if the largest of the three distances between them equals the sum of the other two.
- **Mid-point**: the mid-point M of the segment joining $S(x_1,y_1)$ and $T(x_2,y_2)$ is

$$M = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$$

- **Circle**: a point lies on a circle with centre C and radius r if its distance from C equals r. It lies inside if the distance is less than r and outside if it is greater.

Example 7.1 Are the points $M(-3,-4)$, $A(0,0)$ and $G(6,8)$ on one straight line?

SOLUTION

$MA=\sqrt{9+16}=5$, $AG=\sqrt{36+64}=10$, $MG=\sqrt{81+144}=15$

$MA+AG=5+10=15=MG$

Yes, the three points are collinear.

Example 7.2 $M(-7,1)$ is the mid-point of $A(3,-4)$ and $B(x,y)$. Find B.

SOLUTION

$\frac{3+x}{2}=-7$ gives $x=-17$, and $\frac{-4+y}{2}=1$ gives $y=6$.

$B=(-17,6)$

Example 7.3 Show that $A(1,-8)$, $B(-4,7)$ and $C(-7,-4)$ lie on a circle with centre at the origin. Does $D(-5,6)$ lie inside, on or outside this circle?

SOLUTION

$OA=\sqrt{1+64}=\sqrt{65}$, $OB=\sqrt{16+49}=\sqrt{65}$, $OC=\sqrt{49+16}=\sqrt{65}$

All three are at the same distance from O, so they lie on a circle of radius $\sqrt{65}$.

$OD=\sqrt{25+36}=\sqrt{61}$, which is less than $\sqrt{65}$.

The radius is $\sqrt{65}$ units, and D lies inside the circle.

8 Common mistakes

- Writing the coordinates in the wrong order. The x-coordinate always comes first.
- Thinking a point on an axis is in a quadrant. It is on the boundary and belongs to none.
- Sign errors with negative coordinates: $-3-(-7)=4$, not -10.
- Forgetting to square before adding in the distance formula, or forgetting the square root at the end.
- Saying three points are collinear because two of the distances are equal. Check that two distances **add up** to the third.